

ADESOLA KAREEM

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PROFESSIONAL SUMMARY

Backend Software Engineer with 2+ years of production Python experience specializing in distributed pipelines, high-throughput data systems, and REST API design. At WebMD, owned the full Athena AI ingestion pipeline from design through production and HIPAA deployments.

SKILLS

Languages: Python (5+ yrs), JavaScript/Node.js, Java

Backend + Infra: FastAPI, REST APIs, Pydantic, RabbitMQ, Docker, Kubernetes, GCP, GitLab CI/CD, JWT, OAuth2

Data: PostgreSQL, pgvector, Redis, MongoDB, Solr, SQL

AI/ML + Ops: LLM routing, Embeddings (OpenAI, Azure, Anthropic, Vertex, Cohere), RAG, FAISS, Presidio, Grafana, Prometheus

WORK EXPERIENCE

WebMD - Software Engineer, Backend

Mar 2025 - Present | Python, FastAPI, RabbitMQ, pgvector, PostgreSQL, Redis, Docker, GCP

- Resolved 6 concurrent performance bottlenecks under a 2M-document load test: async locking, TCP reuse, GC safety, bulk CTE deletes, connection cleanup, and logger crash fix
- Reduced database round-trips to 3 for any batch size via bulk `unnest()` inserts (delete + doc + chunk/embedding in one transaction)
- Eliminated pod restarts under load by integrating tenacity retry logic (3 attempts, exponential backoff 1s-8s) for transient 502/503/504 errors
- Built pgvector storage backend from scratch: asyncpg, HNSW index, schema-per-business-unit multi-tenant routing, and GIN full-text index
- Designed RabbitMQ batch pipeline: in-memory buffer with configurable flush on `BATCH_SIZE` or interval; idempotent dedup across 6 composite fields

Nomba - Software Engineer

Nov 2023 - Feb 2024 | Python, MongoDB, Redis, GCP

- Reduced GCP infrastructure costs by 25% through caching strategy and service-level optimization
- Built a high-volume CSV export pipeline backed by MongoDB aggregation and Redis caching

Precious Cornerstone University (NYSC) - Research Assistant

Mar 2024 - Jan 2025

- Trained 20+ faculty on AI tools (ChatGPT, Claude), improving team productivity by ~40%

EDUCATION

Obafemi Awolowo University, Ile-Ife, Nigeria | Aug 2023

BSc Computer Science with Mathematics - First Class Honours (GPA: 4.62/5.0)

CERTIFICATIONS

HackerRank: Java (2024), REST API (2023), Problem Solving (2023), Python (2022)

PROJECTS

Scalable Notification Service | May 2026 - Present

FastAPI, RabbitMQ, Redis, PostgreSQL, Docker, Prometheus, Grafana | github.com/kareemadesola/scalable-notification-service

- Production-grade multi-channel system (email, SMS, push, WebSocket) with dead-letter queues, Redis rate limiting, and ~17K notifications/sec throughput

AI Investigation and Research Assistant | Apr 2026

Python, FAISS, RAG, Agentic pipeline | github.com/kareemadesola/ai-investigation-assistant

- Agentic RAG system combining FAISS vector search, multi-step reasoning, and a hallucination detection layer